



Yuriy Sazontov

Lead / Principal Data & AI Architect

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LinkedIn: linkedin.com/in/sazontov

Portfolio: github.com/sazontov/portfolio/wiki

Professional Summary

Lead / Principal Data & AI Architect with more than 20 years of experience across data, analytics, integration, enterprise systems.

I design platforms from business requirements, system context through C4/HLA, component architecture, physical architecture, data models, integrations, security, sizing, CI/CD, DataOps/MLOps, production rollout.

Core areas: DWH, Lakehouse, Data Vault 2.0, batch/CDC/streaming, metadata management, lineage, data quality, MDM/Profile 360, ML/MLOps, GenAI, Search & RAG, MCP, AI assistants, AI agents, multi-agent systems.

Industry experience covers healthcare, banking, cybersecurity, oil, mining, government, digital media. Delivery experience covers architecture leadership, team management, PoC/MVP execution, hiring, stakeholder alignment, production launches.

Portfolio: [vendor-neutral reference architecture for an enterprise Data, ML & AI Platform](#)

Experience

Freelance

Lead / Principal Data & AI Architect, Independent Consultant

August 2025 — Present · Moscow · Remote

Data, ML, AI architecture consulting for companies across healthcare, banking, cybersecurity.

Healthcare Company Project

- Conducted an AI maturity assessment:
 - assessed architecture, development processes, engineering methods against corporate standards, best practices;
 - identified architectural, organizational, technological gaps, constraints, risks;
 - defined target approaches, organizational measures, architectural measures, implementation roadmap.
- Designed the Digital Twin data mart architecture:
 - defined requirements, use cases, business processes, non-functional requirements, data flows, integrations;
 - developed a C4 architecture for unified patient, physician, healthcare organization profiles;
 - designed a historical analytical profile, online operational profile, low-latency marts, file storage, ML/AI computations;
 - designed change synchronization between analytical, operational environments.
- Reverse-engineered DataHub:
 - developed, implemented an extended C4 architecture for DataHub;
 - designed a microservice interaction graph, lineage, call chains based on contracts, logs, Kubernetes metadata, Kafka metadata;
 - developed, implemented modules for automated service metadata collection;
 - synchronized metadata with the corporate Architecture as Code environment.
- Evaluated adoption of the proprietary Maestro AI framework:
 - analyzed business use cases, expected value, functional constraints, risks;

- assessed compliance with corporate architecture, development, security, operational standards;
- defined the framework's position within the corporate Data/AI Platform, responsibility boundaries, integrations, data flows;
- evaluated the target architecture, prepared recommendations, defined adoption prerequisites.
- Designed data warehouse pipelines for a healthcare organization:
 - designed end-to-end data ingestion, landing, historization, transformation, publication processes;
 - covered batch ingestion, CDC, API integrations, event sources, file sources;
 - defined orchestration, data quality, metadata management, data mart generation environments.
- Designed a secure Kubernetes platform:
 - defined component placement across DMZ, Main environments;
 - designed network isolation, ingress/egress flows, access control, hardening;
 - compared single-cluster, separated-cluster options based on security, integrations, operational risks.
- Designed additional application, integration solutions:
 - Streamlit-based rapid application prototyping, hypothesis validation;
 - Marketplace integration;
 - data access through DBClient.

Independent Reference Model

- Designed, published a vendor-neutral reference architecture for an enterprise Data, ML & AI Platform:
 - performed C4 decomposition from system context, containers through components, integrations, data flows, runtime environments;
 - designed Access, Data Delivery, AI, Data Marts, Orchestration, Ingestion, Landing, Data Core, Data & AI Lab, Metadata, Observability layers;
 - integrated batch, CDC, streaming, API, file pipelines, Data Vault 2.0, lakehouse, analytical marts, operational marts, graph storage, search storage, file storage;
 - designed ML/MLOps, GenAI, Search & RAG, MCP, AI assistant, AI agent, multi-agent system architectures;
 - addressed metadata management, lineage, security, access control, secrets management, consent management, anonymization, logging, monitoring.
- Designed a universal Profile 360 model:
 - designed profiles for people, organizations combining identities, roles, states, relationships, history, documents, events, features, forecasts, interventions, outcomes;
 - combined Customer 360, Client 360, Digital Twin capabilities;
 - designed historical, operational, streaming, search, graph, file, AI-oriented profile projections.
- Designed AI agent, assistant architectures:
 - separated personal AI agents, application assistants, enterprise AI agents, Evolution Agents, multi-agent teams;
 - designed isolated per-user pods, private state, memory, session context, skills, MCP capabilities for personal agents;
 - designed conversational scenarios, embedded business-application scenarios, access to enterprise data, RAG, computations, application operations for assistants;
 - designed task definition, decomposition, delegation, role interaction, result exchange, verifiable artifact creation for enterprise agents;
 - addressed agent identity, role models, permissions, constraints, audit, Human-in-the-Loop approvals, escalations.
- Designed n8n-based AI Automation Runtime, Agentic Assistants Runtime:
 - designed production AI pipelines, governed automations, embedded AI agents, assistant execution;
 - covered API, event, schedule triggers, workflow execution, state persistence, execution history, error handling, approvals, escalations;
 - used Supabase, PostgreSQL for state, context, configuration, execution results;
 - designed LLM integration through LiteLLM, access to enterprise data, services through MCP capabilities, REST API, Kafka;
 - defined boundaries among workflow automation, application assistants, personal AI agents, enterprise AI agents.

- Designed MCP, A2A interaction architecture:
 - designed FastMCP Gateway, specialized MCP servers, discoverable MCP capabilities;
 - developed authentication, authorization based on JWT, RBAC, Keycloak roles, attributes;
 - designed declarative pre-validation, post-validation chains for MCP capability calls;
 - linked validation depth to operation risk, security policies, invocation context;
 - addressed auditing, session management, task delegation, inter-agent result exchange;
 - designed A2A interaction for enterprise agents, Evolution Agents using Hermes, Paperclip.
- Designed the AI Enterprise architecture:
 - created an organizational model comprising one company, 11 teams, 77 AI agent roles;
 - formalized goals, roles, projects, tasks, skills, permissions, budgets, approvals, runs, costs, verifiable artifacts;
 - defined responsibilities across Data Engineering, Data Governance, AI Engineering, MLOps, architecture, operations, cybersecurity;
 - designed runtime environments for personal agents, assistants, enterprise agents, Evolution Agents, agent teams.
- Developed an Architecture as Code approach:
 - created a unified catalog of architecture objects, relationships, technologies, descriptions;
 - formalized hierarchy, traceability, integrity rules, change control;
 - automated quality checks, diagram-description synchronization, Russian documentation, English documentation;
 - established a foundation for relational, graph representations of the architecture.
- Created a public bilingual architecture portfolio:
 - developed C1, C2, C3 models;
 - created a component catalog, architecture decision descriptions;
 - developed a healthcare industry case;
 - published the documentation in the [Architecture Portfolio](#).

Core technologies: Kubernetes, Kafka, Debezium, MinIO/S3, Iceberg, Trino, ClickHouse, PostgreSQL, Redis, OpenSearch, Neo4j, Dagster, dbt, dlt, DataHub, Ranger, ClearML, AutoGluon, FastMCP, n8n, Supabase, LiteLLM, Hermes, Paperclip, Envoy, Keycloak, Vault.

SimbirSoft

Company website: <https://www.simbirsoft.com/en/>

Lead Architect

February 2024 — August 2025 · Moscow · Remote

- Worked on a commercial project, defined the project roadmap, domain model, terminology graph.
- Refined customer requirements, formulated questions, proposed component solutions.
- Designed conceptual, component, physical architecture for a cybersecurity event store, BI platform based on ClickHouse, Kafka, MinIO, Envoy, PostgreSQL, Go, Python.
- Produced use case, sequence, deployment, resource diagrams, built an Excel resource calculator.
- Designed the ClickHouse physical data model:
 - reduced storage dimensions from 1,000 to 200 attributes while supporting extension with an unlimited number of attributes;
 - developed DDL, mapping generators for MergeTree, Kafka Engine, Distributed tables, materialized views, TTL, keys, partitioning, sampling, virtual attributes, Nullable, compression codecs, row policies;
 - analyzed, validated the logical data model, identified inconsistencies using Jupyter Notebooks.
- Designed PostgreSQL metadata database models for BI, ETL Mapping using jsonb, JSON Schema:
 - produced detailed physical ER models in Toad Data Modeler;
 - developed, implemented role-based access to metadata through PostgreSQL functions, data through ClickHouse row policies, authentication, authorization through Envoy, JWT;
 - implemented full-text search across text, jsonb using tsvector;

- installed databases, developed PL/pgSQL functions, procedures, triggers, constraints, initial datasets, migrations, dumps.
- Served as team lead:
 - helped build the team, interviewed candidates, advised on architecture, analytics;
 - worked with customers, explained, aligned complex concepts, architecture, helped define system requirements;
 - managed MVP preparation, implementation, production rollout.

Kaspersky

Company website: <https://www.kaspersky.com/>

Data Platform Architect

November 2022 — February 2024 · Moscow

- Conducted analysis for a secure data platform across cloud, on-premises environments:
 - collected, analyzed requirements;
 - evaluated available Open Source solutions against requirements, constraints;
 - developed CI/CD, secure DataOps, secure MLOps process models;
 - developed PoC scenarios, prepared data for platform research.
- Designed the data platform architecture:
 - conceptual architecture, HLA/C4 for cloud, on-premises environments;
 - component, physical architecture for PoC environments;
 - technology-stack selection, justification;
 - sizing, research planning, environment delivery support, research scenario implementation.

Core stack: Kubernetes, operators for stateful components, Astra Linux, ClickHouse, Greenplum, PostgreSQL, MinIO, Dremio, ArangoDB, Airflow, Superset, Jenkins, Camunda, HashiCorp Vault, Keycloak, NATS, Benthos, DataHub.

Accenture

Company website: <https://www.accenture.com/>

Data Architecture Manager

December 2021 — April 2022 · Moscow

Consulting for a Major Mining Company, Kazakhstan

- Selected the technology stack.
- Advised on implementation of the development process, evolution of the enterprise data model.
- Supported baseline sizing.
- Developed process models for core data warehouse use cases.

ML Project Leadership for a Bank, Russia — NLP for HR

- Managed day-to-day project delivery in Asana, led a five-person team.
- Led data extraction, preparation, EDA, model development, model application, testing.
- Led delivery of a PoC scoring model matching résumés to target roles, predicting potential career transitions.

Stack: doc2vec, word2vec, BERT, Keras, TensorFlow.

Nedra

Company website: <https://nedra.digital/>

Data Platform Architect

March 2021 — December 2021 · Moscow · Remote

- Designed a data warehouse architecture from scratch for a major oil-industry customer.
- Produced component, information, domain, process, sequence, use case, data models.
- Developed the detailed conceptual warehouse model: layers, functional subsystems, information model, core structures, transformation flows, standard business services, entity naming, attribute naming, flow naming.

- Detailed PXF virtualization, CDC replication, data masking, data-request processing architectures.

Stack: Greenplum, Hadoop, Debezium, Kafka, Airflow, NiFi, Dremio, Hasura.

Alfa-Bank

Company website: <https://www.alfabank.com/>

Chief BI Architect (DWH/Hadoop)

February 2019 — March 2020 · Moscow

- Contributed to the customer-side DWH/BI technology strategy.
- Developed, aligned, approved architecture decisions.
- Designed solutions based on Hadoop, Oracle, Informatica PowerCenter.

Sber

Company website: <https://www.sberbank.com/>

Chief Data Engineer (CDE), Corporate Banking

August 2017 — January 2019 · Moscow

Managed data mart development, support, ETL for Client Analytics, Sales Campaigns within Corporate Banking.

Results:

- established the Data Engineering team;
- designed, implemented the Data Engineering process;
- delivered more than 10 data marts;
- organized support for more than 20 data marts.

Stack: Teradata, SAS, Oracle, Pentaho, Java.

Parma Technologies Group

Company website: <https://parma.ru/>

Deputy Head of Technology Department

April 2016 — August 2017 · Moscow

- Selected, assembled an alternative company technology stack based on Open Source products.
- Led a research team, environment preparation, presentations, demonstrations.
- Designed information system architectures, researched, tested Open Source software.
- Participated in customer negotiations as architect, prepared technical tender sections, organized source-system research.
- Recruited, trained specialists.

Technologies: Linux, Docker, WSO2, Hadoop, Oracle, PostgreSQL, MySQL, MongoDB, ClickHouse, MonetDB, Pentaho, Jasper, ODI, PHP, Java, C.

Prognoz

Deputy Head of Design, Development & Testing Center, Software Architect

February 2013 — April 2016 · Moscow

- Planned delivery, recruited, allocated resources, conducted interviews, formed teams; managed more than 100 specialists during peak periods.
- Handled administrative responsibilities, served as architect, project manager, technical director.
- Provided technical leadership for major government, enterprise portal projects using data, analytics.
- Designed 3NF, Kimball, Data Vault 2.0 data models, UML software architecture, IDEF0/BPMN/YAWL processes, physical architecture, functional architecture.
- Conducted technical assessments, reverse engineering, reengineering of existing customer information systems, advised customers, participated in government stakeholder negotiations.
- Solved specialized architecture issues, coordinated mitigation of DDoS attacks, consequences.

Designed, delivered more than ten portal solutions across these system classes:

- regional, federal public portals;
- government information systems supporting business-process modeling, execution;
- government information systems for regulatory reporting, analytical reporting;
- geographic information systems;
- integration solutions using SMEV, Public Services Portal, electronic signatures, other government services.

Architecture, analysis responsibilities:

- business requirements, derived system requirements;
- UML diagrams using design patterns;
- physical, logical Oracle ER models, detailed PL/SQL layers, database reverse engineering, Oracle option selection;
- detailed IDEF0 business-process models, BPMN/YAWL workflow models;
- physical, functional, role, organizational architectures, access models, competency models;
- domain models, formal logical controls, business rules, validation rules;
- web-service catalogs, web-client catalogs;
- competency requirements, high-level project plans;
- work with extensive Russian-language, English-language technical documentation.

Additional technical practice:

- software development, testing, mentoring, leadership of developers, testers;
- unit testing, load testing, coding standards, quality-management standards, Code Review;
- EJB, JPA, Servlet, JSP, JSTL, JSF, JAX-WS, JAX-RS, JNDI, JMS, JTA, JAAS, JavaMail, JCA, CDI, core language constructs, design patterns;
- database administration, performance tuning for servers, clusters, databases using EM, TOAD, AWR, Oracle system views;
- deployment of EAR, JAR, WAR packages, configuration of WebLogic, Coherence, GoldenGate, ODI, RCU;
- Windows, Solaris, Linux.

Sostav.RU

Company website: <https://www.sostav.ru/>

Chief Technology Officer (CTO)

March 2004 — December 2012 · Moscow

Responsibilities:

- led development, launch of internet portals, services;
- defined portal concepts, business models, business processes, operating standards, business metrics;
- planned projects, assigned work, controlled timelines, quality;
- managed day-to-day operations for a seven-person technical team;
- organized system administration, technical support, user feedback, incident resolution, performance optimization;
- supported editorial teams in Russia, Ukraine, content management, commercial operations, advertising placement, campaign reporting, custom projects.

Results — designed, launched 10 portals, services:

- **Sostav.RU** — Russian internet portal covering advertising, marketing, PR.
- **Sostav.UA** — Ukrainian internet portal covering advertising, marketing, PR.
- **ForumSostav.RU** — professional marketing forum.
- **Videosostav.RU** — business video-hosting service with more than 80,000 videos.
- **TopSostav.RU** — social network for advertising, marketing professionals.
- **TenSostav.RU** — advertising-market tender platform.
- **Sostav.RU/blogs** — business blogs on Sostav.RU.

- **GlosSostav.RU** — marketing terminology dictionary.
- **Sostav.RU/shop** — online store on Sostav.RU.
- **DepotWPF.RU** — website for the Depot WPF branding agency.

Education

University of Artificial Intelligence

2021 · Data Science, Neural Networks, Machine Learning & Artificial Intelligence

Russian Presidential Academy of National Economy & Public Administration, Moscow

2010 · Economics, MBA CIO

Kemerovo State University, Kemerovo

1997 · Physics

Professional Development & Certifications

- **2020** — **Big Data**, University of California San Diego, Coursera
- **2020** — **Advanced Machine Learning**, National Research University Higher School of Economics, Coursera
- **2020** — **Python Programming**, Moscow Institute of Physics & Technology, Mail.Ru Group, E-Learning Development Fund, Coursera
- **2020** — **Machine Learning in Finance**, Sberbank Corporate University, Coursera
- **2019** — **Machine Learning & Data Analysis**, Moscow Institute of Physics & Technology, Yandex, E-Learning Development Fund, Coursera
- **2019** — **Deep Learning & Neural Networks**, SkillFactory
- **2018** — **Teradata**, Teradata
- **2007** — **Oracle 11g**, Specialist Training Center at Bauman Moscow State Technical University

Skills

Architecture: Enterprise Architecture, Solution Architecture, Data Architecture, IT Architecture, C4, HLA, UML, BPMN, IDEF0, Architecture as Code, DWH design, Lakehouse design, Data Vault 2.0.

Data & Integration: Data Engineering, Data Modeling, BI, Big Data, DWH, ETL/ELT, batch, CDC, streaming, API, Data Quality, Data Governance, Metadata Management, Data Lineage, MDM, Profile 360, SQL, Oracle, PostgreSQL, ClickHouse.

ML & AI: Data Science, Machine Learning, Neural Networks, NLP, Text Processing, Keras, TensorFlow, MLOps, GenAI, Search & RAG, MCP, AI Assistants, AI Agents, Multi-Agent Systems.

Leadership: Project Management, IT Leadership, Engineering Team Leadership, Team Building, Interviewing, Stakeholder Management, PoC/MVP Delivery, Production Rollout.

Core Technologies: Kubernetes, Kafka, Debezium, MinIO/S3, Iceberg, Trino, ClickHouse, PostgreSQL, Redis, OpenSearch, Neo4j, Dagster, dbt, dlt, DataHub, Ranger, ClearML, AutoGluon, FastMCP, n8n, Supabase, LiteLLM, Hermes, Paperclip, Envoy, Keycloak, Vault, Python, SQL.

Languages: Russian — Native; English — B2, Upper Intermediate.